Using the MaizeGDB genome browser to evaluate gene model annotation quality and gain functional insights

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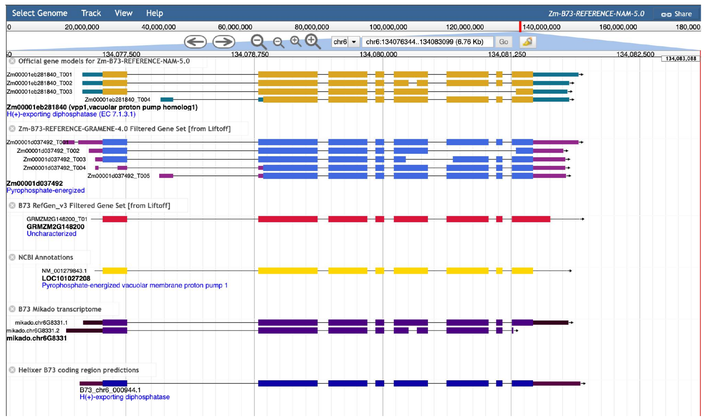
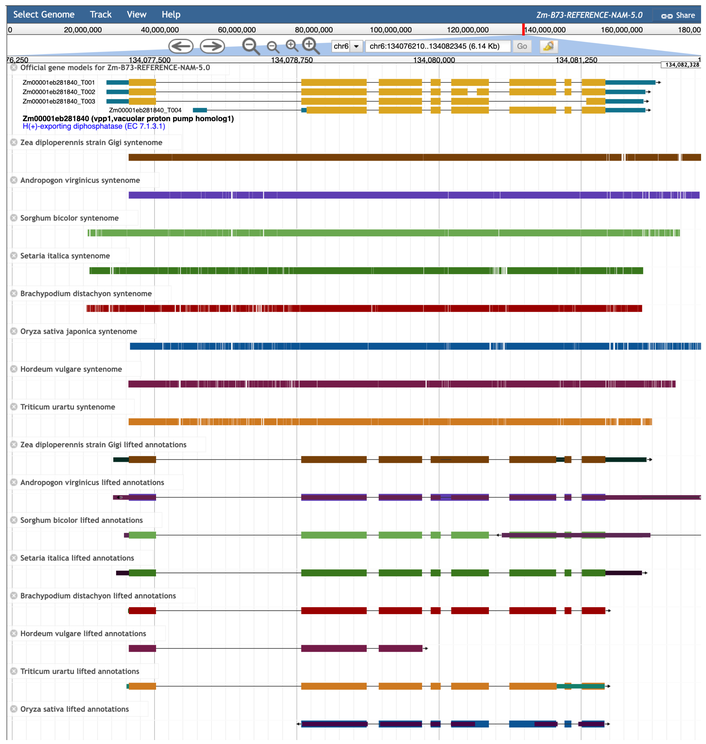
Gene Model Annotations



The B73_v5 genome browser includes the official **gene model annotations**, along with gene models mapped from B73_v4 and B73_v3 using three different methods. Additional alternative annotations are available from GenBank, as well as from the Mikado[1] and Helixer[2] annotation pipelines.

1 doi: 10.1093/gigascience/giy093
2 doi: 10.1093/bioinformatics/btaa1044

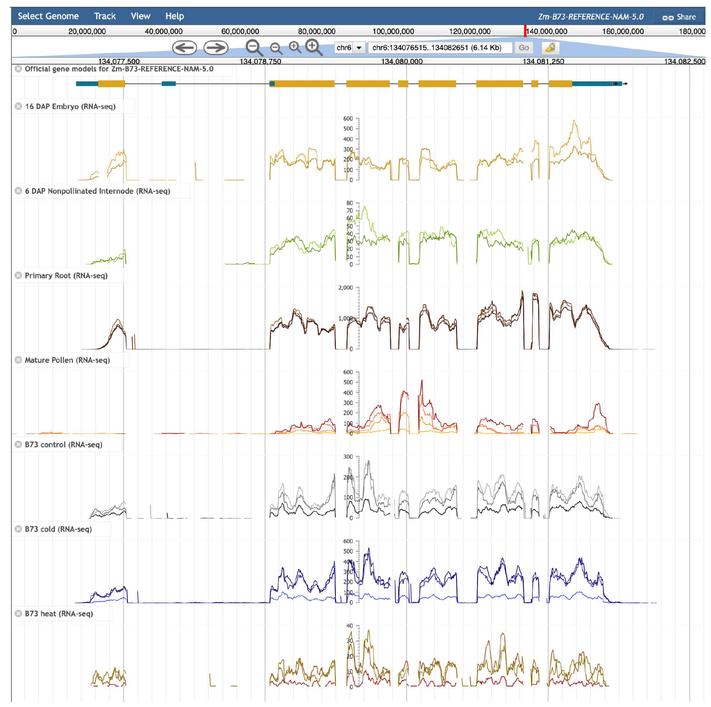
Syntenome



Syntenome refers to genomic regions in maize B73 version 5 that are predicted to be syntenic with genomic regions in other plants. These tracks display syntenic alignment coordinates between maize and the indicated plant species as determined by the program *AnchorWave*[1]. Gene model annotations were lifted from each genome to B73v5 using the program *Liftoff*[2]. This process was performed separately for each B73v5 chromosome FASTA file.

1 doi: 10.1073/pnas.2113075119
2 doi: 10.1093/bioinformatics/btaa1016

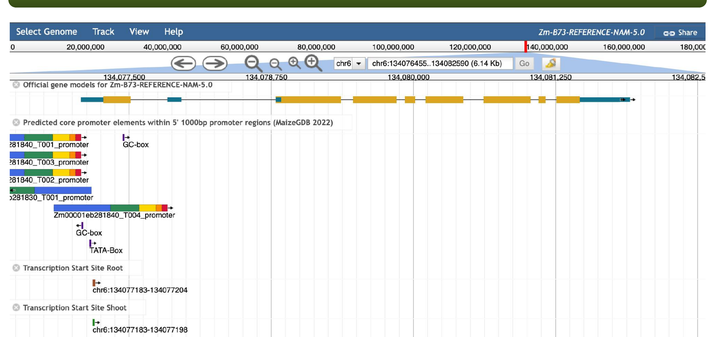
Gene Expression



MaizeGDB has curated and mapped over 300 **gene expression** datasets from seven projects, including the NAM consortium[1], the Stelpflug 2016[2] tissue atlas, the Walley 2016[3] mRNA and protein abundance dataset, and the recent Hayford 2024[4] dataset, which includes biotic and abiotic stress conditions from 24 studies.

1 doi: 10.1126/science.abg5289
2 doi: 10.3835/plantgenome2015.04.0025
3 doi: 10.1126/science.aag1125
4 doi: 10.1186/s12864-024-10443-7

Other tracks



MaizeGDB features over 1000 **tracks** that empower researchers to evaluate gene model annotations and gain functional insights into maize genetics and genomics. Additional tracks highlighted here include predicted core elements and transcription start site locations.

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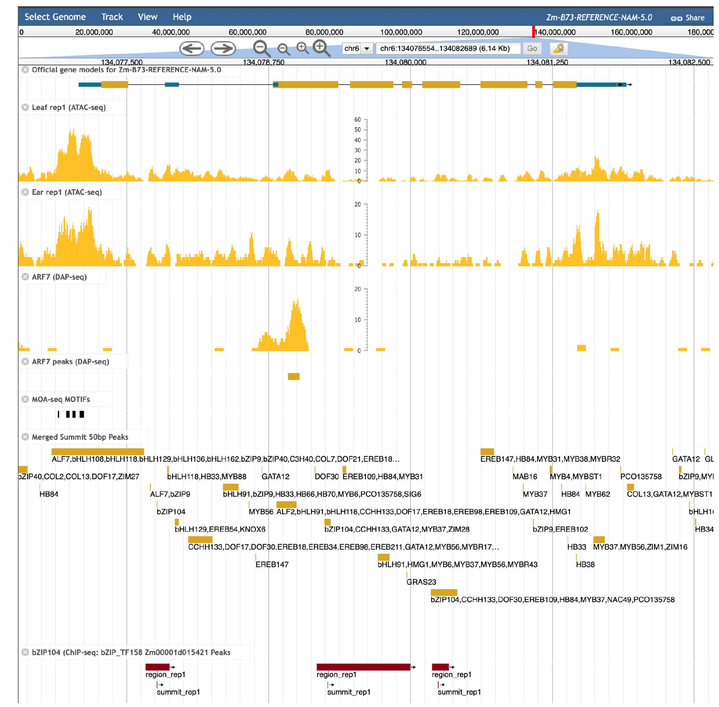
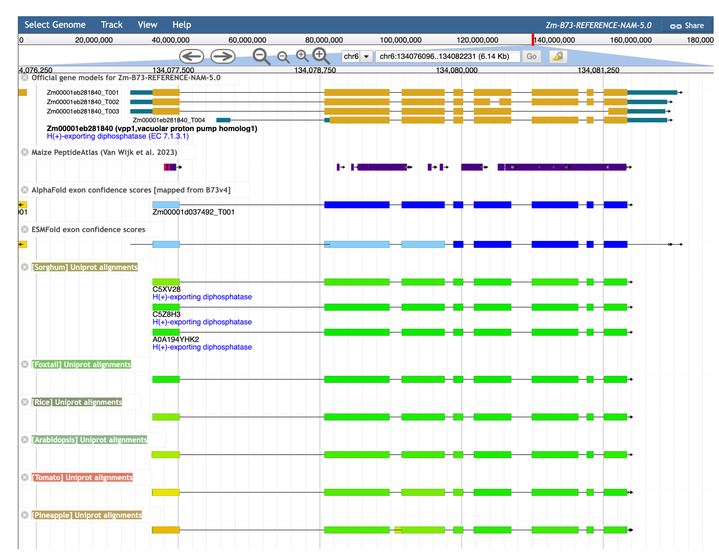
Epigenetics and DNA Binding



MaizeGDB offers over 600 **epigenomics and DNA-binding** datasets on its genome browser. Data tracks include open chromatin regions, transcription factor binding sites, enhancer elements, differential methylation patterns, and histone modifications.

Proteomics



Proteomics tracks available on the genome browser include aligned protein fragments from the Maize PeptideAtlas[1], predicted protein structure quality scores (averaged per exon) using AlphaFold[2] and ESMFold[3], and alignments of over 20 proteomes using Miniprot[4] from UniProt and Phytozome.

1 doi: 10.1021/acs.jproteome.4c00320
2 doi: 10.1038/s41586-021-03819-2
3 doi: 10.1126/science.ade2574
4 doi: 10.1093/bioinformatics/btad014